ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the covered half remains brown with iodine	Light needed.
2	**(c)** light energy to chemical energy	Stored in glucose.
3	**(b)** nitrogen-fixing bacteria	Rhizobium.
4	**(b)** chlorophyll is needed	Pigment.
5	**(b)** chemical energy to usable energy	Released.
6	**(c)** nitrogen left by pulses	Rhizobium.
7	**(d)** NaOH absorbs carbon dioxide	No CO2.
8	**(d)** more glucose than used in respiration	Growth.
9	**(c)** mushrooms have no chlorophyll and absorb food from dead matter	Fungi are saprotrophs.
10	**(b)** increases with light intensity	Closer lamp.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
12	Lack of oxygen (1); They cannot respire (1).	1+1
13	Spines and thick cuticle (1); Fewer stomata (1).	1+1
14	No light for photosynthesis (1); Stored starch was used (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
16	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
17	Faster photosynthesis (1); More oxygen (1); Gas escapes (1).	1+1+1
	OR: Starch increases through the day (1); Photosynthesis exceeds respiration (1); Falls at night (1).	1+1+1
18	They absorb carbon dioxide (1); They release oxygen (1); They trap dust on their leaves (1).	1+1+1
	OR: Large in shade (1); Small or spines in desert (1); Floating in ponds (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vary lamp distance (1); Count bubbles per minute (1); Keep temperature and CO2 same (1); Table and graph (1); Rate rises with light (1).	1+1+1+1+1
	OR: Set up three conditions (1); Same plant type (1); Starch test after some days (1); Only suitable one forms starch (1); Conclusions (1).	1+1+1+1+1
20	Parasite or saprotroph (1); Check connection to living tissue (1); Observe on dead wood (1); Test for starch (1); Conclude (1).	1+1+1+1+1
	OR: Photosynthesis takes CO2, gives O2 (1); Respiration reverse (1); Balance in nature (1); Burning adds CO2 (1); Plants reduce it (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it gains mass and grows	Balanced gas exchange.
2	**(c)** saprotroph	Feeds on bread.
3	**(d)** high humidity	Humid air reduces.
4	**(a)** photosynthesis is less than respiration	Low light.
5	**(d)** Cuscuta, a parasite	Lack chlorophyll.
6	**(a)** a plant with many large leaves	More surface for water loss.
7	**(c)** nitrogen or magnesium	Chlorophyll needs them.
8	**(a)** insectivorous	Needs nitrogen.
9	**(d)** stops making food and starves	CO_2 runs out.
10	**(b)** they need to capture light under tall canopy	Light is limited.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dissolves chlorophyll (1); Makes colour change clearer (1).	1+1
12	Alga provides food (1); Fungus provides water and protection (1).	1+1
13	Soil is poor in nitrogen (1); They trap insects for nitrogen (1).	1+1
14	Photosynthesis is faster than respiration (1); Net release of oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Raise photosynthesis (1); Faster growth (1); Higher yields (1).	1+1+1
16	Fixes nitrogen to nitrates (1); Plant makes amino acids (1); Proteins formed (1).	1+1+1
17	First in food chain (1); Make food from sunlight (1); Support consumers (1).	1+1+1
	OR: Magnesium needed for chlorophyll (1); Less chlorophyll (1); Pale leaves (1).	1+1+1
18	They respond to light (1); More light for food making (1); Plants respond to stimuli (1).	1+1+1
	OR: Place a stem in coloured water (1); Colour rises through xylem (1); Cut section shows coloured vessels (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Leaf makes food (1); Phloem transports (1); Potato tuber stores starch (1); Carrot root stores (1); Used for next season (1).	1+1+1+1+1
	OR: Plants respire all the time (1); Photosynthesis only in light (1); Both occur in day (1); Respiration continues at night (1); Statement false (1).	1+1+1+1+1
20	Less photosynthesis (1); Less transpiration so less rain (1); More CO_2 (1); Tree planting (1); Green belts (1).	1+1+1+1+1
	OR: Modified leaves (1); Digestive juices (1); They still photosynthesise (1); Need extra nitrogen (1); Rooted, have chlorophyll (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** magnesium	Magnesium is part of chlorophyll.
2	**(a)** oxygen	Gas.
3	**(b)** thick cuticle	Waxy cuticle.
4	**(a)** green plants	Producers.
5	**(a)** alcohol	They are sensitive to pollutants.
6	**(a)** the green stem	Leaves are spines.
7	**(a)** the Sun	Light energy.
8	**(c)** roots absorb dissolved minerals and need oxygen	Respiration needed.
9	**(b)** minerals dissolved in water are pulled up	Transport in xylem.
10	**(c)** photosynthesis	Uptake.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adds nitrogen (1); Improves soil (1).	1+1
12	They grow towards light (1); More light reaches the leaves for making food (1).	1+1
13	Dust blocks light and stomata (1); Less photosynthesis (1).	1+1
14	Reduce water loss (1); Protected from direct sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Leaves catch light and gases (1); Root hairs absorb water (1); Large area helps both (1).	1+1+1
16	Close stomata (1); Prevent water loss (1); No photosynthesis at night (1).	1+1+1
17	Starch in the cotyledons (1); Used for growth (1); Until leaves make food (1).	1+1+1
	OR: After dark: no starch (1); After light: starch present (1); Photosynthesis restarts (1).	1+1+1
18	Respiration cannot occur (1); No photosynthesis (1); Plant dies (1).	1+1+1
	OR: Legume with Rhizobium (1); Lichen with fungus (1); Parasites on hosts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Evaporation at leaves (1); Suction pulls water in xylem (1); Continuous column (1); Roots absorb to replace (1); Minerals travel (1).	1+1+1+1+1
	OR: Rhizobium lives in legume roots (1); It fixes nitrogen for the crop (1); Fertiliser use is reduced (1); Soil becomes fertile (1); Farming becomes cheaper (1).	1+1+1+1+1
20	Adequate sunlight (1); Crop rotation with legumes (1); Irrigation scheduling (1); Spacing of plants (1); Mulching (1).	1+1+1+1+1
	OR: Sunny spot (1); Compost (1); Pulses in rotation (1); Mulch to reduce transpiration (1); Regular watering (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.